

AI system for early detection and treatment of diseases

Artificial intelligence (AI) will revolutionise healthcare. With the help of machine learning, ECG, EEG, or X-ray images will be accurately analysed for early detection of diseases. However, implanting AI within the human body is a major technical challenge. TU Dresden scientists have now developed a bio-compatible implantable AI platform that classifies in real time healthy and pathological patterns in biological signals such as heartbeats. It detects pathological changes even without medical supervision. In trials, the AI was able to differentiate between healthy heartbeats from three common arrhythmias with an 88 % accuracy rate.

Artificial polymer-based neural network. The strongly nonlinear behavior of these networks enables their use in reservoir computing (Credit: <https://tu-dresden.de/tu-dresden/newsportal>)



AI based weather forecasting system for farms



Researchers figured a way to utilise the best and the most accurate climate models for weather forecasting in smart farms (Credit: www.electronicsforu.com)

Researchers at the James Cook University (JCU) working on smart irrigation systems have developed a way to choose the most accurate weather forecast out of those offered in the week leading up to a given day. The forecasting is done based on climate models to produce multiple forecasts for the same rainfall event. To achieve this, a hybrid climate learning model has been developed, which learns from a combination of the climate model data and provides crucial answers to climate-related questions. First, a probability based network evaluates multiple forecasts for different rainfall patterns. Then a deep-learning neural network reprocesses the forecasts to produce a better prediction for the next day.

Phase-change memory cells for use in flexible electronics

Among the most promising data storage technologies is the phase-change memory, which is thousands of times faster than conventional hard drives but is not the most

energy-efficient among emerging memory types. Now, Stanford University engineers have overcome a key obstacle that has limited widespread adoption of phase-change